

Data Mining — 10-Page Master Quick Revision

High-Yield Formula Sheet, Schema Matrices & Top BIT Mesra PYQ Solutions

🔗 10-Page Master Quick Revision — Data Mining (CS24303)

Page 1-2: Attribute Types, Statistical Summaries & Distance Metrics

Page 5-6: OLAP Cubes, Star vs. Snowflake Schemas & Operations

Page 9-10: Antimonotonic Constraints, Colossal Patterns & Exam PYQs

Page 3-4: Data Cleaning, Chi-Square Correlation & Normalization Formulas

Page 7-8: Apriori Join/Prune, FP-Tree Construction & Lift Analysis

⚡ High-Yield Data Mining Formulas Matrix

Concept	Exact Formula / Rule	Key Insight
Chi-Square (χ^2)	$\chi^2 = \sum \frac{(O-E)^2}{E}$ where $E = \frac{\text{row_total} \times \text{col_total}}{N}$	Tests nominal attribute independence.
Min-Max Norm	$v' = \frac{v - \min}{\max - \min} (\text{new_max} - \text{new_min}) + \text{new_min}$	Maps to target range linearly.
Z-Score Norm	$v' = \frac{v - \mu}{\sigma}$	Zero mean, unit variance.
Apriori Property	Subset of frequent itemset is frequent $\implies$ Superset of infrequent is infrequent	Foundation of candidate pruning.
Lift Metric	$\text{Lift}(A, B) = \frac{P(A \cup B)}{P(A)P(B)}$	> 1 : Positively correlated; < 1 : Negatively correlated.